

Chapter 10: Analysing data

8 lessons · one lesson for each exercise · each lesson is one hour

NSW syllabus outcome

Stage 4 Statistics: MA4-DAT-C-01 classifies and displays data using a variety of graphical representations; MA4-DAT-C-02 analyses simple datasets using measures of centre, range and shape of the data.

CHAPTER

10

10.01 Interpreting graphs

10.02 Misleading graphs

10.03 Dot plots

10.04 Stem-and-leaf plots

10.05 Mean and mode

10.06 Median and range

10.07 Analysing dot plots and stem-and-leaf plots

10.08 Comparing data

How to use this booklet

1 Each lesson is **one hour**. Start with the **learning intentions** and **success criteria**, write the **key terms** and use the **notes** box.

WE DO **Grey boxes.** Your teacher works each example (parts a, b, c) on the board. **Copy every step** into the space.

2 Every lesson has three levels: **Easy**, then **Medium**, then **Challenging**. Work down the page: the questions get harder as you go.

YOU DO **Blue boxes.** Now it is your turn. Work on your own. Quick drill answers go on the line; show working in the boxes.

DRILL **Skill drill pages.** After the Easy and the Medium examples there is a full page of short practice questions. Work down the page, write answers on the lines, then mark it and record your score and time.

3 Finished early? Try the **extension**. At the end, **show off your skill** on the tear-off exit ticket and hand it in.

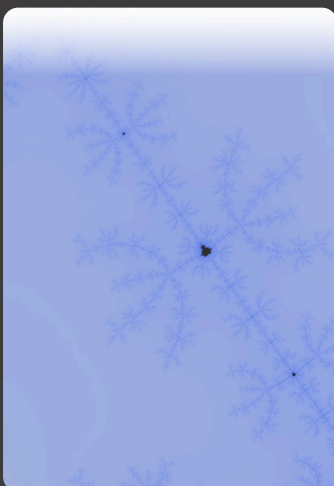
4 Missed a lesson? Each lesson ends with a **summary** with a worked example for every level. Start there.

Contents and progress tracker

Lesson	Topic	Page	Date	Done
10.01	Interpreting graphs	3		☐
10.02	Misleading graphs	9		☐
10.03	Dot plots	15		☐
10.04	Stem-and-leaf plots	21		☐
10.05	Mean and mode	27		☐
10.06	Median and range	33		☐
10.07	Analysing dot plots and stem-and-leaf plots	39		☐
10.08	Comparing data	45		☐

In this chapter you will

- interpret picture, column, sector and line graphs
- identify misleading graphs
- read and draw dot plots and stem-and-leaf plots
- find the mean, mode, median and range
- analyse data shown in plots
- compare two sets of data



About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \rightarrow z^2 + c$, thousands of times for every point. Every booklet shows a different part of the set: this one is snowflake-like branches between the main body and the top bulb. Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

LESSON
10.01

Interpreting graphs

Date _____

Learning intentions

We are learning to:

- read and interpret picture, sector, column and line graphs

Success criteria

I can:

- ☐ read the title, axes, key and scale first
- ☐ read values from a graph
- ☐ answer questions and explain trends

Key terms Write the definition as your teacher explains it.

Key

Scale

Sector graph

Notes

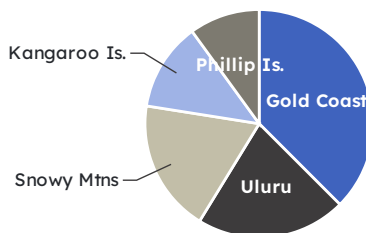
Level 1 · Getting started

Easy · 26 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Use the sector graph.

Favourite holiday destinations (80 people)



a What is the most popular destination?

Your teacher works this out. Copy every step.

b Which destination did 17 people prefer?

Your teacher works this out. Copy every step.

YOU DO

On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **26 QUESTIONS**

REMEMBER Read the title, labels and key.

ROUND 1

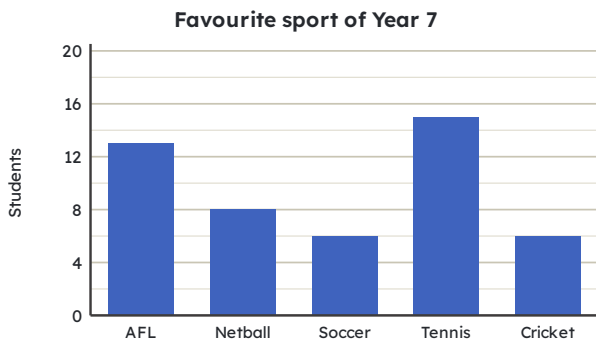
Questions 1–12: reading scales: a column ends exactly halfway between two gridlines. What value does it show?

- | | | | |
|---------------------------------------|--------------------------------------|--|---------------------------------------|
| 1 between 45 and 50
_____ | 2 between 30 and 35
_____ | 3 between 10 and 12
_____ | 4 between 160 and 180
_____ |
| 5 between 140 and 160
_____ | 6 between 14 and 16
_____ | 7 between 35 and 40
_____ | 8 between 50 and 60
_____ |
| 9 between 80 and 90
_____ | 10 between 16 and 20
_____ | 11 between 120 and 140
_____ | 12 between 6 and 8
_____ |

YOU DO On your own: **Easy** keep going: the next rounds.

14 QUESTIONS MORE

ROUND 2 Questions 13–20: use the column graph. Read the scale carefully.



13 How many chose AFL?

14 How many chose Netball?

15 How many chose Soccer?

16 How many chose Tennis?

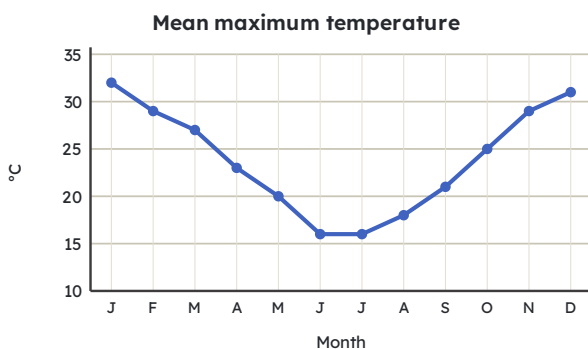
17 How many chose Cricket?

18 Which sport was most popular?

19 How many students were surveyed?

20 How many more chose Tennis than AFL?

ROUND 3 Questions 21–26: use the line graph.



21 What was the temperature in March?

22 What was the temperature in June?

23 Which month was hottest?

24 Which month was coldest?

25 By how much did it change from March to June?

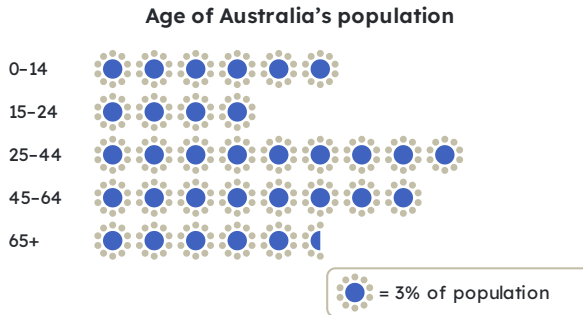
26 What is the range of the temperatures?

Level 2 · Building up

Medium · 12 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Use the picture graph.



a Which age group has the smallest percentage?

Your teacher works this out. Copy every step.

b What percentage are aged 15-24?

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

12 QUESTIONS

REMEMBER Check the scale on each axis.

ROUND 1 Questions 1-12: a survey result: write it as a percentage.

1 17 of 50 students

2 135 of 200 students

3 19 of 20 students

4 24 of 25 students

5 8 of 50 students

6 29 of 40 students

7 7 of 20 students

8 20 of 25 students

9 114 of 200 students

10 21 of 25 students

11 4 of 40 students

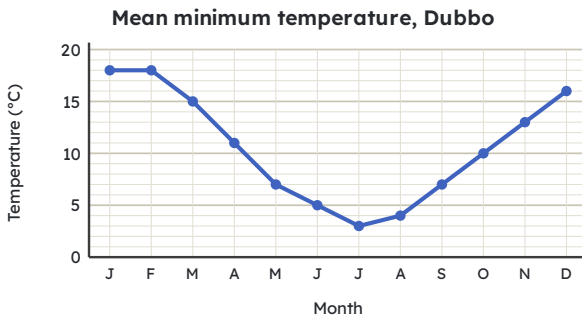
12 11 of 20 students

Level 3 · Challenge

Challenging · 3 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Use the line graph.



- a** Which months have the highest mean minimum?
What is it?

Your teacher works this out. Copy every step.

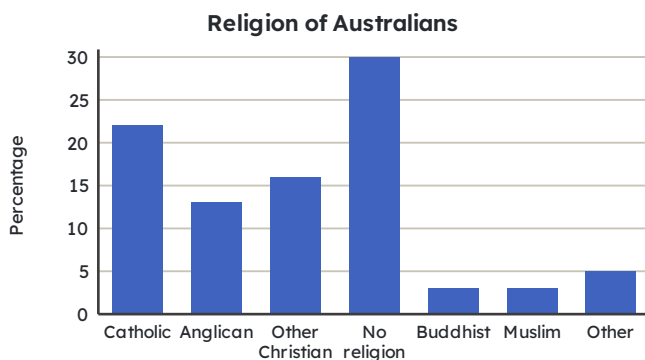
- b** Describe the trend from January to July.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Use the column graph.

3 QUESTIONS

REMEMBER Read values carefully with a ruler.



- 1** What percentage are Catholic?

- 2** Which religion is followed by 13%?

- 3** What is the most common response?



Extension · Stretch yourself

Finished the challenge? Stretch yourself here.



Worked example

In the sector graph, what angle is the Gold Coast sector?

1. Gold Coast is 30 out of 80 people
2. $\frac{30}{80} \times 360^\circ$

Answer: 135°

YOU DO

On your own: read the worked example, then try these.

- E1** Find the angle of the Uluru sector (1 d.p.).

- E2** Draw a divided bar graph 16 cm long for the holiday data. How long is each part?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

In the sector graph, which destination was least popular?

Level 2

In the line graph, what is the mean minimum in April?

Level 3

In the column graph, how many more % have no religion than are Catholic?

Lesson summary

Missed the lesson? Start here.

Remember

1. Read the title, labels and key.
2. Check the scale on each axis.
3. Read values carefully with a ruler.
4. Describe trends: increasing, decreasing, highest, lowest.

Level 1

Most popular (sector)

1. Largest sector

Answer: Gold Coast

Level 2

15–24 (picture graph)

1. 4 symbols $\times$ 3%

Answer: 12%

Level 3

Highest mean minimum

1. Highest points

Answer: Jan and Feb,
18°C

Extension

Gold Coast angle

1. $30/80 \times 360$

Answer: 135°

My reflection: what did I learn today? Which question was hardest, and why?

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LESSON
10.02

Misleading graphs

Date _____

Learning intentions

We are learning to:

- identify features that make a graph misleading

Success criteria

I can:

- ☐ check whether the vertical axis starts at zero
- ☐ check the scale is even and the graph has labels
- ☐ redraw a misleading graph correctly

Key terms Write the definition as your teacher explains it.

Misleading

Broken axis

Notes

Level 1 · Getting started

Easy · 12 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Use the line graph.



a What is incorrect about it?

Your teacher works this out. Copy every step.

b What misleading impression does it give?

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. 12 QUESTIONS

REMEMBER Check the vertical axis starts at 0 (or shows a break).

ROUND 1 Questions 1–4: misleading (M) or fair (F)?

1 The sectors add to 130%.

2 There is a break mark (⚡) shown on the axis.

3 The scale has equal steps.

4 The pictures get bigger in two directions.

YOU DO

On your own: **Easy** keep going: Round 2.

8 QUESTIONS MORE

**ROUND
2**

Questions 5–12: what makes the graph misleading? Choose: scale, axis start, picture size, or missing labels.

5 The columns start at 80 so small changes look huge.

6 The larger bag of money is drawn 3 times as wide.

7 There is no title or key.

8 The y-axis goes 0, 5, 10, 50, 100.

9 A picture twice as tall is also twice as wide.

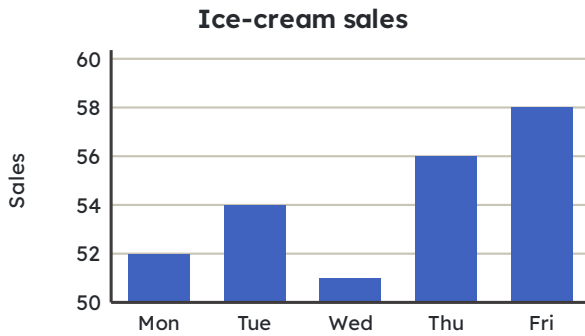
10 The y-axis starts at 1000.

11 The axes have no titles.

12 The steps on the axis are not equal.

Level 2 • Building up

Medium • 12 questions

WE DO With your teacher: copy what they write on the board.**Example 2** Use the column graph.

- a** Thursday sales look double Wednesday's. Is that true?

Your teacher works this out. Copy every step.

- b** What are the actual sales on Wednesday and Thursday?

Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line. **12 QUESTIONS****REMEMBER** Check the scale goes up in equal steps.**ROUND 1** Questions 1–6: explain why each graph could mislead a reader.

- 1** A graph of test scores has an axis from 60 to 70.

- 2** A company shows only the months when profits rose.

- 3** A line graph squashes the time axis to make a rise look steep.

- 4** Pictures of houses: one house is drawn twice as tall and twice as wide.

- 5** A sector graph is tilted in 3D.

- 6** Sales went from 96 to 98. The axis starts at 95.

YOU DO On your own: **Medium** keep going: Round 2.

6 QUESTIONS MORE

**ROUND
2**

Questions 7–12: the axis starts at the number given. How many times taller does column B look than column A? How many times bigger is it really? (1 d.p.)

7 A = 51, B = 54, axis starts at 50

8 A = 72, B = 80, axis starts at 70

9 A = 105, B = 110, axis starts at 100

10 A = 22, B = 30, axis starts at 20

11 A = 11, B = 14, axis starts at 10

12 A = 92, B = 96, axis starts at 90

Level 3 · Challenge

Challenging · 3 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Think about misleading graphs.

a Name 3 other ways a graph can mislead.

Your teacher works this out. Copy every step.

b Why would a company want a misleading graph?

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Think it through.

3 QUESTIONS

REMEMBER Check there is a title and labelled axes.

1 A graph uses pictures of bags of money; the 2019 bag is twice as tall and twice as wide as the 2014 bag. Why is this misleading?

2 A graph has no title and no labels on the axes. Why is this a problem?

3 A column graph has a scale 0, 10, 20, 50, 100. What is wrong?



Extension · Stretch yourself

Finished the challenge? Stretch yourself here.



Worked example

Profit grew from \$2.2 million to \$4.0 million. By what percentage did it grow?

1. Increase = 1.8 million
2. $\frac{1.8}{2.2} \times 100\%$

Answer: $\approx 81.8\%$

YOU DO

On your own: read the worked example, then try these.

- E1 On the misleading graph, the 2019 point looks about 8 times higher than 2014. Explain why.

- E2 Draw a graph of the same data that makes growth look very small.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Give one reason a graph might be misleading.

Level 2

A column graph's axis starts at 90. Why is that a problem?

Level 3

How should the company profits graph be fixed?

Lesson summary

Missed the lesson? Start here.

Remember

1. Check the vertical axis starts at 0 (or shows a break).
2. Check the scale goes up in equal steps.
3. Check there is a title and labelled axes.
4. Watch for pictures that change in 2 directions.

Level 1

Profit graph

1. Axis starts at 2

Answer: growth looks bigger

Level 2

Sales Wed vs Thu

1. 51 vs 56

Answer: not double

Level 3

Uneven scale 0, 10, 20, 50

1. Steps not equal

Answer: misleading

Extension

Growth %

1. $1.8 \div 2.2$

Answer: $\approx 81.8\%$

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
10.03

Dot plots

Date _____

Learning intentions

We are learning to:

- read and draw dot plots

Success criteria

I can:

- ☐ read the frequency of each value
- ☐ find the mode and any outlier
- ☐ draw a dot plot from data

Key terms Write the definition as your teacher explains it.

Dot plot

Outlier

Mode

Notes

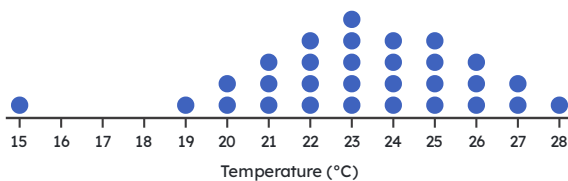
Level 1 · Getting started

Easy · 15 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Use the dot plot.

Maximum temperatures at Bega in April



a What is the mode?

Your teacher works this out. Copy every step.

b What is the outlier?

Your teacher works this out. Copy every step.

YOU DO

On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line.

15 QUESTIONS

REMEMBER Each dot is one piece of data.

ROUND 1 Questions 1–9: find the mode (the most common value).

1 1, 9, 8, 4, 1, 7, 2

2 8, 2, 2, 5, 6, 1, 4

3 8, 8, 3, 7, 3, 7, 8

4 7, 6, 8, 1, 9, 2, 6

5 6, 6, 5, 9, 5, 6, 9

6 9, 8, 5, 6, 8, 3, 7

7 2, 5, 7, 7, 6, 4, 8

8 5, 3, 6, 2, 2, 4, 1

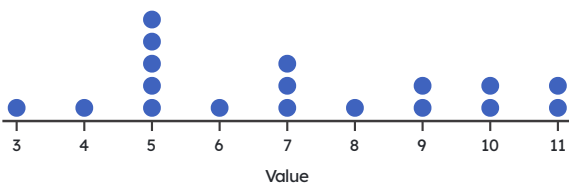
9 8, 2, 4, 3, 7, 1, 7

YOU DO On your own: **Easy** keep going: Round 2.

6 QUESTIONS MORE

ROUND 2 Questions 10–15: use the dot plot.

Pets owned by families



10 How many values are there?

11 What is the mode?

12 What is the range?

13 What is the smallest value?

14 What is the largest value?

15 How many values are greater than 7?

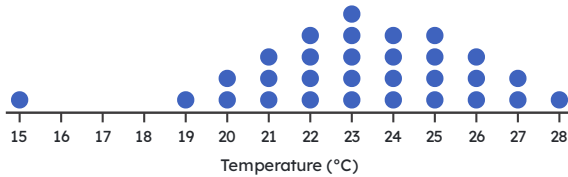
Level 2 · Building up

Medium · 16 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Use the dot plot.

Maximum temperatures at Bega in April



a On how many days was it 25°C?

Your teacher works this out. Copy every step.

b On what fraction of days was it below 20°C?

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

16 QUESTIONS

REMEMBER Mode: the tallest column.

ROUND 1 Questions 1–16: find the range (largest – smallest).

1 4, 34, 2, 4, 15, 19, 38

2 6, 5, 26, 33, 7, 36, 37

3 13, 38, 7, 28, 35, 11, 29

4 12, 21, 10, 30, 28, 35, 29

5 26, 23, 12, 37, 34, 40, 11

6 14, 39, 40, 24, 23, 24, 23

7 8, 35, 13, 20, 4, 25, 32

8 17, 20, 15, 11, 7, 19, 9

9 35, 31, 2, 25, 20, 16, 18

10 38, 19, 29, 4, 7, 18, 26

11 31, 11, 35, 26, 28, 11, 20

12 5, 21, 32, 17, 38, 35, 21

13 21, 31, 17, 2, 39, 19, 14

14 27, 8, 38, 20, 14, 8, 9

15 3, 23, 5, 17, 14, 12, 25

16 4, 15, 15, 8, 13, 5, 33



Level 3 · Challenge

Challenging · 2 questions · then try the extension

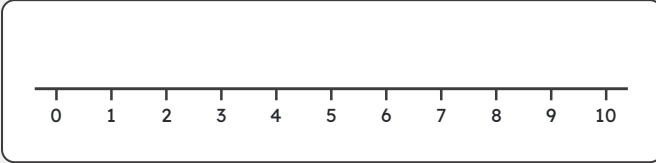
WE DO

With your teacher: copy what they write on the board.

Example 3

Draw a dot plot, then describe.

- a Draw a dot plot: 2, 3, 3, 4, 4, 4, 5, 6, 6, 9



- b Describe the shape of the Set A plot.

Your teacher works this out. Copy every step.

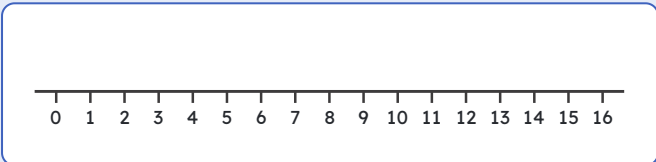
YOU DO

On your own: **Challenging** Draw and analyse.

2 QUESTIONS

REMEMBER Outlier: far away from the rest.

- 1 Minutes late: 0, 2, 1, 0, 3, 0, 15, 1, 2, 0, 1, 0. Draw a dot plot.



- 2 For Question 1, find the mode and the outlier. How does the outlier affect the data?



Extension · Stretch yourself

Finished the challenge? Stretch yourself here.



Worked example

What percentage of April days at Bega were 25°C or more?

- Count: $25 (4) + 26 (3) + 27 (2) + 28 (1) = 10$
- $\frac{10}{30} \times 100\%$

Answer: 33.3%

YOU DO

On your own: read the worked example, then try these.

- E1 What percentage of games had no goals? (Set B)

- E2 Add a game with 3 goals to Set B. What is the new mode?

YOU DO

Show off your skill

Choose ONE question. Show your working.

Name _____ Class _____

Level 1

What is a dot plot?

Level 2

In Set B, what is the mode?

Level 3

Draw a dot plot for 1, 2, 2, 3, 3, 3, 8. Which value is the outlier?

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Each dot is one piece of data.
 2. Mode: the tallest column.
 3. Outlier: far away from the rest.
 4. Draw: a number line with an even scale, one dot per value.

Level 1

Mode of Bega plot

1. Tallest column

Answer: 23°C

Level 2

Outlier

1. Far from the rest

Answer: 15°C

Level 3

Below 20°C

1. 2 days out of 30

Answer: $\frac{1}{15}$

Extension

25°C or more

1. 10 of 30 days

Answer: 33.3%

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
10.04

Stem-and-leaf plots

Date _____

Learning intentions

We are learning to:

- read and draw ordered stem-and-leaf plots

Success criteria

I can:

- ☐ read values from a stem-and-leaf plot using the key
- ☐ order the leaves in each row
- ☐ draw a stem-and-leaf plot from data

Key terms Write the definition as your teacher explains it.

Stem

Leaf

Key

Notes

Level 1 · Getting started

Easy · 14 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Use the stem-and-leaf plot.

Masses of football players (kg), unordered

Stem	Leaf
7	9 5 8 1 6 9 4
8	2 5 0 0 8 0 3 2 4
9	0 8 4 2 2 0
10	5 6 4

Key: 7 | 9 = 79 kg

a How many players are there?

Your teacher works this out. Copy every step.

b What is the lowest mass?

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **14 QUESTIONS**

REMEMBER The stem is the tens digit; the leaf is the ones digit.

ROUND 1 Questions 1–8: write the value shown (key 3 | 4 = 34).

1 4 | 7

2 1 | 3

3 1 | 1

4 3 | 9

5 1 | 2

6 6 | 5

7 2 | 7

8 7 | 9

YOU DO On your own: **Easy** keep going: Round 2.

6 QUESTIONS MORE

ROUND 2 Questions 9–14: use the stem-and-leaf plot.

9 How many scores are there?

10 What is the lowest score?

Test scores

Stem	Leaf
2	1 8 9
3	2 2 9
4	9
5	2 4 4 9
6	0 2 3 4 5

Key: 2 | 7 = 27

11 What is the highest score?

12 How many scores are in the 30s?

13 How many scores are 50 or more?

14 What is the range?

Level 2 · Building up

Medium · 12 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Use the stem-and-leaf plot.

Masses of football players (kg), unordered

Stem	Leaf
7	9 5 8 1 6 9 4
8	2 5 0 0 8 0 3 2 4
9	0 8 4 2 2 0
10	5 6 4

Key: 7 | 9 = 79 kg

a Which mass occurs most often?

Your teacher works this out. Copy every step.

b What is the highest mass?

Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line. **12 QUESTIONS**

REMEMBER Always include a key, e.g. 7 | 9 = 79.

ROUND 1 Questions 1–12: write the values in the row (key 3 | 4 = 34).

1 1 | 2 5 5 8

2 3 | 1 1 3 8

3 7 | 4 4 8 9

4 2 | 1 4 6 9

5 5 | 0 1 2 6

6 1 | 5 5 8 9

7 3 | 0 1 4 6

8 1 | 1 2 4 4

9 5 | 5 5 8 9

10 8 | 0 2 8 9

11 6 | 0 2 4 5

12 9 | 0 4 5 9



Level 3 · Challenge

Challenging · 2 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Draw an ordered plot, then explain.

a Draw an ordered stem-and-leaf plot for Set A.

Stem	Leaf
7	
8	
9	
10	

Key: ____ | ____ = _____

b Why do we order the leaves?

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Draw and analyse.

2 QUESTIONS

REMEMBER Order the leaves from smallest to largest.

1 Order the Set A plot (masses of players).

Stem	Leaf
7	
8	
9	
10	

Key: ____ | ____ = _____

2 For the ordered plot, how many players weigh 90 kg or more? What fraction is that?



Extension · Stretch yourself

Finished the challenge? Stretch yourself here.



Worked example

Write the data 3.4, 3.7, 4.1, 4.5, 4.5, 5.0, 5.2 as a stem-and-leaf plot.

1. Stems are the whole numbers, leaves the tenths
2. Key: $3 \mid 4 = 3.4$

Answer: $3 \mid 4 \ 7; 4 \mid 1 \ 5 \ 5; 5 \mid 0 \ 2$

YOU DO

On your own: read the worked example, then try these.

- E1** Draw a stem-and-leaf plot for 105, 112, 118, 121, 125, 130 (stems 10, 11, 12, 13).

- E2** What would a key of $2 \mid 5 = 0.25$ mean?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

In the key $6 \mid 3 = 63$, what is the stem?

Level 2

Write the values in the row $4 \mid 0 \ 2 \ 7$.

Level 3

Draw an ordered stem-and-leaf plot for 12, 25, 17, 21, 14, 29.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. The stem is the tens digit; the leaf is the ones digit.
 2. Always include a key, e.g. $7 | 9 = 79$.
 3. Order the leaves from smallest to largest.
 4. Line the leaves up in columns so the plot shows the shape.

Level 1

Players

1. Count the leaves

Answer: 25

Level 2

Lowest mass

1. First leaf of first stem

Answer: 71 kg

Level 3

Ordered plot

1. Sort each row

Answer: $7 | 1\ 4\ 5\ 6\ 8\ 9\ 9;$
 $8 | 0\ 0\ 0\ 2\ 2\ 3\ 4\ 5\ 8; 9 | 0$
 $0\ 2\ 2\ 4\ 8; 10 | 4\ 5\ 6$

Extension

Decimals

1. Stem = whole number

Answer: $3 | 4\ 7; 4 | 1\ 5\ 5;$
 $5 | 0\ 2$

My reflection: what did I learn today? Which question was hardest, and why?

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LESSON
10.05

Mean and mode

Date _____

Learning intentions

We are learning to:

- find the mean and mode of a set of data

Success criteria

I can:

- ☐ find the mean: add the values, divide by how many
- ☐ find the mode: the most common value
- ☐ round the mean to 2 decimal places

Key terms Write the definition as your teacher explains it.

Mean

Mode

Notes

Level 1 · Getting started

Easy · 25 questions

WE DO With your teacher: copy what they write on the board.**Example 1** Find the mean and the mode.**a** Find the mean of 4, 3, 2, 5, 6, 4, 4.

Your teacher works this out. Copy every step.

b Find the mode of 4, 3, 2, 5, 6, 4, 4.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **25 QUESTIONS****REMEMBER** Mean = sum of values ÷ number of values.**ROUND 1** Questions 1–9: find the mean: add the values, then divide by how many there are.**1** 6, 1, 2**2** 9, 16, 8**3** 11, 3, 8, 2**4** 17, 9, 9, 5**5** 10, 11, 14, 5**6** 11, 7, 3**7** 14, 18, 13**8** 9, 6, 16, 7, 7**9** 20, 12, 8, 6, 4

YOU DO

On your own: **Easy** keep going: Round 2.

16 QUESTIONS MORE

ROUND 2

Questions 10–25: find the mode.

10 8, 1, 12, 12, 11, 12

11 10, 11, 11, 1, 8, 2

12 8, 11, 12, 11, 9, 2

13 11, 4, 4, 8, 6, 2

14 9, 1, 4, 10, 10, 12

15 5, 3, 11, 12, 12, 10

16 2, 6, 12, 10, 1, 12

17 4, 3, 11, 4, 10, 9

18 12, 10, 2, 12, 11, 7

19 11, 11, 10, 6, 3, 12

20 1, 5, 7, 9, 3, 1

21 3, 3, 7, 3, 7, 2

22 10, 1, 10, 6, 12, 4

23 11, 6, 9, 9, 5, 12

24 10, 5, 1, 2, 4, 1

25 4, 10, 5, 8, 8, 7

 Level 2 · Building up

Medium · 20 questions

WE DO With your teacher: copy what they write on the board.**Example 2** Find the mean and the mode.**a** Find the mean of 6, 12, 11, 12, 10, 6, 6, 10, 6.

Your teacher works this out. Copy every step.

b Find the mode of 6, 12, 11, 12, 10, 6, 6, 10, 6.

Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line. **20 QUESTIONS****REMEMBER** Mode = the value that occurs most often.**ROUND 1** Questions 1–12: find the mean to 1 decimal place.**1** 49, 38, 33, 10, 27**2** 42, 7, 12, 16**3** 27, 27, 43, 30, 14, 37, 37**4** 27, 15, 6, 50, 39, 31, 25**5** 37, 23, 19, 30**6** 42, 29, 43, 40**7** 17, 41, 17, 44, 31, 29, 18**8** 6, 45, 16, 34, 14**9** 44, 29, 13, 44, 21, 50**10** 41, 25, 42, 16, 49, 12, 18**11** 9, 9, 19, 12**12** 40, 46, 36, 25, 36, 16, 25

YOU DO

On your own: **Medium** keep going: Round 2.

8 QUESTIONS MORE

ROUND 2

Questions 13–20: the mean is given. Find the missing value.

13 2, 11, 3, ? (mean 14)

14 11, 2, 1, 33, ? (mean 20)

15 36, 25, 22, 1, ? (mean 18)

16 20, 7, 17, ? (mean 17)

17 22, 16, 22, ? (mean 20)

18 8, 1, 3, 5, ? (mean 5)

19 3, 10, 3, ? (mean 6)

20 28, 6, ? (mean 19)

Level 3 • Challenge

Challenging · 3 questions · then try the extension

WE DO

With your teacher: copy what they write on the board.

Example 3

Find the mean and mode, then solve.

- a** Find the mean and mode of 8, 4, 1, 1, 4, 1, 3, 6.

Your teacher works this out. Copy every step.

- b** The mean of 4 numbers is 7. Three are 5, 8 and 9. Find the fourth.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Challenging** Problems.

3 QUESTIONS

REMEMBER There can be more than one mode, or none.

- 1 The mean of 4 numbers is 7. Three are 5, 8 and 9. Find the fourth.

[illegible]

- 2 Sam's test marks are 72, 85, 68 and 91. What must he score on the next test for a mean of 80?

[illegible]

- 3 Write 5 numbers with a mean of 10 and a mode of 8.

[illegible]

Extension · Stretch yourself

Finished the challenge? Stretch yourself here.

Worked example

A class of 25 has a mean mark of 64. A new student scores 90. Find the new mean.

1. Total = $25 \times 64 = 1600$
2. New total 1690, 26 students
3. $1690 \div 26$

Answer: 65

YOU DO

On your own: read the worked example, then try these.

- E1 The mean of 6 numbers is 12. One number, 20, is removed. Find the new mean.

- E2 Explain why the mean does not have to be one of the data values.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO

Show off your skill

Choose ONE question. Show your working.

Name _____ Class _____

Level 1

Find the mode: 5, 8, 5, 3, 5, 8.

Level 2

Find the mean: 4, 6, 8, 10, 12.

Level 3

Find the mean (2 d.p.): 3, 7, 7, 9.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Mean = sum of values $\div$ number of values.
 2. Mode = the value that occurs most often.
 3. There can be more than one mode, or none.
 4. Round the mean to 2 decimal places.

Level 1

4, 3, 2, 5, 6, 4, 4

1. sum 28, 7 values

Answer: mean 4

Level 2

4, 3, 2, 5, 6, 4, 4

1. 4 occurs 3 times

Answer: mode 4

Level 3

8, 4, 1, 1, 4, 1, 3, 6

1. sum $28 \div 8$

Answer: mean 3.5,
mode 1

Extension

New student

1. $1690 \div 26$

Answer: 65

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
10.06

Median and range

Date _____

Learning intentions

We are learning to:

- find the median and range of a set of data

Success criteria

I can:

- ☐ order the data before finding the median
- ☐ find the middle value, or the mean of the two middle values
- ☐ find the range: highest – lowest

Key terms Write the definition as your teacher explains it.

Median

Range

Notes

Level 1 · Getting started

Easy · 25 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Find the median and the range.

a Find the median of 4, 3, 2, 5, 6, 4, 4.

b Find the range of 4, 3, 2, 5, 6, 4, 4.

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **25 QUESTIONS**

REMEMBER Order the data first.

ROUND 1 Questions 1–9: find the median: order the values, then find the middle one.

1 6, 10, 23, 8, 23

2 11, 12, 14, 13, 9

3 25, 9, 15, 26, 10

4 4, 12, 20, 6, 20

5 11, 1, 21, 22, 9

6 22, 26, 30, 13, 2

7 18, 11, 11, 11, 7

8 10, 16, 28, 27, 20

9 26, 3, 13, 13, 29

YOU DO

On your own: **Easy** keep going: Round 2.

16 QUESTIONS MORE

ROUND 2

Questions 10–25: find the range.

10 33, 29, 27, 36, 46

11 21, 49, 3, 15, 32

12 1, 20, 29, 25, 48

13 43, 27, 19, 16, 19

14 46, 28, 5, 50, 22

15 31, 27, 8, 13, 2

16 45, 4, 33, 17, 33

17 16, 21, 24, 23, 45

18 27, 38, 39, 8, 4

19 45, 47, 23, 34, 46

20 48, 20, 20, 33, 16

21 45, 24, 1, 3, 22

22 13, 25, 42, 7, 41

23 35, 35, 46, 5, 10

24 41, 32, 5, 12, 4

25 38, 15, 48, 2, 37

Level 2 · Building up

Medium · 27 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Find the median and the range.

a Find the median of 6, 12, 11, 12, 10, 6, 6, 10, 6.

Your teacher works this out. Copy every step.

b Find the range of 6, 12, 11, 12, 10, 6, 6, 10, 6.

Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line. **27 QUESTIONS**

REMEMBER Median = the middle value.

ROUND 1 Questions 1–15: find the median. There is an even number of values: take the middle two and halve their sum.

1 35, 26, 7, 21

2 1, 35, 23, 23

3 23, 26, 6, 36, 11, 12

4 18, 39, 24, 1, 22, 35

5 12, 17, 13, 26, 8, 19

6 38, 12, 30, 3

7 13, 36, 19, 17

8 17, 3, 12, 27

9 11, 25, 18, 13, 20, 32

10 24, 8, 19, 17, 9, 2

11 17, 39, 20, 37

12 18, 5, 21, 39, 12, 2

13 31, 2, 22, 34

14 6, 40, 1, 35

15 2, 5, 9, 7

YOU DO On your own: **Medium** keep going: Round 2.

12 QUESTIONS MORE

ROUND 2 Questions 16–27: find the median and the range.

16 57, 21, 14, 25, 47, 22, 21

17 24, 24, 22, 48, 13, 37, 10, 21

18 44, 29, 21, 18, 30, 58

19 27, 57, 33, 44, 41, 28, 57, 27

20 43, 15, 47, 47, 17, 43

21 60, 17, 10, 33, 50, 46, 48, 52

22 45, 25, 54, 32, 55, 52, 45, 33

23 33, 29, 53, 42, 15, 13, 50, 37

24 46, 29, 43, 33, 27

25 29, 26, 52, 43, 57, 52

26 48, 50, 28, 29, 35, 53, 41

27 39, 25, 36, 19, 14, 55, 37, 60

Level 3 · Challenge

Challenging · 3 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Find the median and range, then compare.

a Find the median and range of 8, 4, 1, 1, 4, 1, 3, 6.

Your teacher works this out. Copy every step.

b Which is affected more by an outlier: the median or the mean?

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Problems.

3 QUESTIONS

REMEMBER Even number of values: median = mean of the middle two.

1 Find the mean, median, mode and range of 2, 3, 3, 4, 5, 25.

2 Which measure (mean or median) best describes the data in Question 1? Why?

3 Write 5 numbers with median 6 and range 10.



Extension · Stretch yourself

Finished the challenge? Stretch yourself here.



Worked example

Five numbers have mean 6, median 5, mode 3 and range 8. Find them.

1. Mode 3 and median 5: 3, 3, 5, ?, ?
2. Sum = 30, so the last two add to 19
3. Range 8: largest = $3 + 8 = 11$, so the other is 8

Answer: 3, 3, 5, 8, 11

YOU DO

On your own: read the worked example, then try these.

- E1** Four numbers have median 7, mode 9 and range 6. Find them (hint: 9 appears twice).

- E2** Why might a company report the median salary instead of the mean?

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Find the median: 6, 2, 9, 4, 7.

Level 2

Find the range: 18, 5, 12, 30.

Level 3

Find the median: 3, 8, 10, 11.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Order the data first.
 2. Median = the middle value.
 3. Even number of values: median = mean of the middle two.
 4. Range = highest – lowest.

Level 1

4, 3, 2, 5, 6, 4, 4

1. Ordered: 2, 3, 4, 4, 4, 5, 6

Answer: median 4

Level 2

6, 12, 11, 12, 10, 6, 6, 10, 6

1. Ordered: 6, 6, 6, 6, 10, 10, 11, 12, 12

Answer: median 10

Level 3

8, 4, 1, 1, 4, 1, 3, 6

1. $8 - 1$

Answer: range 7

Extension

mean 6, median 5,
mode 3, range 8

1. Build the list

Answer: 3, 3, 5, 8, 11

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
10.07

Analysing dot plots and stem-and-leaf plots

Date _____

Learning intentions

We are learning to:

- find the range, median, mode and mean from a dot plot or stem-and-leaf plot

Success criteria

I can:

- ☐ list or count the data values from the plot
- ☐ use the order of the plot to find the median
- ☐ calculate the mean and range

Key terms Write the definition as your teacher explains it.

Frequency

Cluster

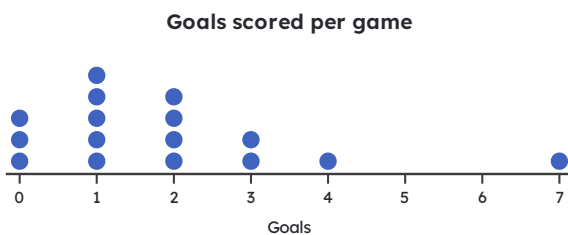
Notes

Level 1 · Getting started

Easy · 9 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Use the dot plot.



a Find the range.

Your teacher works this out. Copy every step.

b Find the median.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line.

9 QUESTIONS

REMEMBER Count the values (n).

ROUND 1 Questions 1–3: find the range of the stem-and-leaf plot.

1 2 | 2 5
3 | 7 7
4 |
5 |
6 | 1 2

2 3 | 9
4 | 3
5 |
6 | 0 4 5 6

3 2 | 4 7
3 | 9
4 | 5 8 9

YOU DO

On your own: **Easy** keep going: Round 2.

6 QUESTIONS MORE

ROUND 2

Questions 4–9: use the stem-and-leaf plot. Give the mean to 1 d.p.

4 How many values?

5 Mode

Heights of seedlings (mm)

Stem	Leaf
3	2 5 8
4	3 5
5	5 7
6	0 1 9
7	2

Key: 4 | 5 = 45 mm

6 Median

7 Range

8 Mean

9 How many values are below 50?

Level 2 · Building up

Medium · 16 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Use the stem-and-leaf plot.

Test scores	
Stem	Leaf
4	5
5	0 0 1 2
6	1 3 4 6 6 6 7 8
7	4 5

Key: 5 | 2 = 52

a Find the mode.

Your teacher works this out. Copy every step.

b Find the median.

Your teacher works this out. Copy every step.

YOU DO

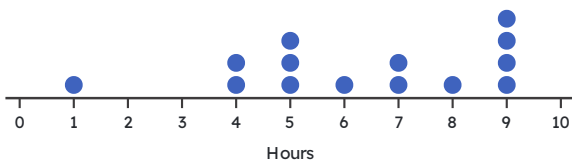
On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

16 QUESTIONS

REMEMBER Range: highest – lowest.

ROUND 1 Questions 1–8: use the dot plot. Give the mean to 1 d.p.

Hours of sleep



1 How many values?

2 Mode

3 Median

4 Range

5 Mean

6 Which value is an outlier?

7 Find the range without the outlier.

8 Does the outlier change the median much?

YOU DO On your own: **Medium** keep going: Round 2.

8 QUESTIONS MORE

ROUND 2 Questions 9–16: use the dot plot. Give the mean to 1 d.p.



9 How many values?

10 Mode

11 Median

12 Range

13 Mean

14 Which value is an outlier?

15 Find the range without the outlier.

16 Does the outlier change the median much?

Level 3 · Challenge

Challenging · 3 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Use the stem-and-leaf plot.

Test scores

Stem	Leaf
4	5
5	0 0 1 2
6	1 3 4 6 6 6 7 8
7	4 5

Key: 5 | 2 = 52

a Set B: find the mean (2 d.p.).

Your teacher works this out. Copy every step.

b Describe where the data in Set B clusters.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Use the stem-and-leaf plot of player masses (ordered).

3 QUESTIONS

REMEMBER Median: the value in position $(n + 1) \div 2$.

Masses of players (kg)

Stem	Leaf
7	1 4 5 6 8 9 9
8	0 0 0 2 2 3 4 5 8
9	0 0 2 2 4 8
10	4 5 6

Key: 7 | 9 = 79 kg

1 Range

2 Median

3 Mode



Extension · Stretch yourself

Finished the challenge? Stretch yourself here.



Worked example

One more player of mass 120 kg joins the team.
How do the mean and median change?

1. Old mean 85.88, median 83
2. New mean 87.19, median 83.5

Answer: the mean rises more than the median

YOU DO

On your own: read the worked example, then try these.

- E1** Which measure would you use to describe the typical player's mass? Why?

- E2** Remove the 7-goal game from the dot plot. Find the new mean.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

A dot plot has dots at 1, 2, 2, 3, 3, 3.
Find the mode.

Level 2

Find the median of 5 | 1 3 4 7 (key 5 | 1 = 51).

Level 3

Find the range of 2 | 3 8, 3 | 0 5, 4 | 1 (key 2 | 3 = 23).

Lesson summary

Missed the lesson? Start here.

Remember

1. Count the values (n).
2. Range: highest – lowest.
3. Median: the value in position $(n + 1) \div 2$.
4. Mean: add all values (count dots $\times$ values) $\div n$.

Level 1

Goals: range

1. $7 - 0$

Answer: 7

Level 2

Goals: median

1. $n = 16$

Answer: 1.5

Level 3

Scores: mean

1. sum $918 \div 15$

Answer: 61.2

Extension

Add a 120 kg player

1. Mean moves more

Answer: mean up more

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
10.08

Comparing data

Date _____

Learning intentions

We are learning to:

- compare two sets of data using plots and statistics

Success criteria

I can:

- ☐ draw a back-to-back stem-and-leaf plot
- ☐ find the mean, median, mode and range for each group
- ☐ compare the groups and decide which is more consistent

Key terms Write the definition as your teacher explains it.

Back-to-back

Consistent

Notes

Level 1 · Getting started

Easy · 16 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Use the back-to-back plot.

Girls	Stem	Boys
8	1	0 7
9 8	2	8 9
9 7 5	3	0 8
8	4	0
9 5 3 2 1	5	0 5 8
9 6 0	6	2 6
6 5 3	7	2 3 3 4 5
3	8	5 8
4	9	3

Key: 3 | 5 = 35

a What does the plot show?

Your teacher works this out. Copy every step.

b What was the highest mark? Who scored it?

Your teacher works this out. Copy every step.

YOU DO

On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line.

16 QUESTIONS

REMEMBER Put the two groups on the two sides of the stems.

ROUND 1 Questions 1–6: which group is more consistent (smaller range)?

1 A: range 26, B: range 27

2 A: range 18, B: range 8

3 A: range 11, B: range 17

4 A: range 25, B: range 22

5 A: range 22, B: range 21

6 A: range 15, B: range 10

YOU DO On your own: **Easy** keep going: Round 2.

10 QUESTIONS MORE

ROUND 2 Questions 7–16: which group has the higher mean?

7 A: 16, 16, 16, 4
B: 13, 12, 9, 9

8 A: 7, 8, 17, 3
B: 5, 2, 20, 2

9 A: 9, 8, 5, 5
B: 13, 6, 2, 20

10 A: 6, 15, 20, 20
B: 3, 2, 12, 6

11 A: 16, 13, 16, 8
B: 5, 12, 10, 11

12 A: 14, 16, 7, 19
B: 7, 12, 13, 11

13 A: 12, 6, 8, 11
B: 10, 3, 19, 13

14 A: 19, 11, 16, 10
B: 19, 18, 18, 7

15 A: 12, 7, 2, 10
B: 13, 2, 3, 3

16 A: 17, 14, 2, 14
B: 2, 9, 20, 14

Level 2 · Building up

Medium · 12 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Use the back-to-back plot.

Girls	Stem	Boys
8	1	0 7
9 8	2	8 9
9 7 5	3	0 8
8	4	0
9 5 3 2 1	5	0 5 8
9 6 0	6	2 6
6 5 3	7	2 3 3 4 5
3	8	5 8
4	9	3

Key: 3 | 5 = 35

a Find the median mark for the girls.

Your teacher works this out. Copy every step.

b Find the range for the boys.

Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line (show working in the boxes). **12 QUESTIONS**

REMEMBER Leaves get larger moving away from the stem.

ROUND 1 Questions 1–6: find the mean and the range of each group. Which group did better? Which is more consistent?

1 Class A: 10, 9, 10, 13, 13
Class B: 8, 6, 13, 14, 15

2 Class A: 7, 7, 7, 20, 18
Class B: 11, 14, 11, 14, 8

3 Class A: 10, 18, 6, 19, 15
Class B: 16, 6, 7, 9, 19

4 Class A: 16, 14, 6, 19, 12
Class B: 19, 18, 12, 11, 8

5 Class A: 11, 9, 8, 5, 13
Class B: 17, 13, 6, 14, 7

6 Class A: 16, 16, 16, 8, 9
Class B: 6, 14, 11, 11, 12

YOU DO On your own: **Medium** keep going: Round 2.

6 QUESTIONS MORE

ROUND 2 Questions 7–12: find the median of each group. Which median is higher?

- 7 A: 49, 55, 49, 51, 13
B: 36, 36, 59, 40, 19

- 8 A: 43, 16, 19, 24, 46
B: 18, 17, 45, 21, 21

- 9 A: 24, 27, 18, 32, 28
B: 28, 44, 18, 21, 40

- 10 A: 46, 28, 28, 23, 59
B: 21, 34, 54, 16, 46

- 11 A: 58, 51, 57, 40, 39
B: 36, 38, 41, 27, 24

- 12 A: 58, 33, 28, 47, 46
B: 43, 23, 27, 49, 57

Level 3 · Challenge

Challenging · 3 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Compare the girls and the boys.

Girls	Stem	Boys
8	1	0 7
9 8	2	8 9
9 7 5	3	0 8
8	4	0
9 5 3 2 1	5	0 5 8
9 6 0	6	2 6
6 5 3	7	2 3 3 4 5
3	8	5 8
4	9	3

Key: 3 | 5 = 35

- a Compare the girls and the boys.

Your teacher works this out. Copy every step.

- b Which group is more consistent? Why?

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Compare and explain.

3 QUESTIONS

REMEMBER Compare centre: mean or median.

- 1 Comment on the differences between the girls and the boys on the assignment.

- 2 Which group was more consistent? Give reasons.

- 3 Which measure (mean or median) would you use to compare these groups? Why?



Extension · Stretch yourself

Finished the challenge? Stretch yourself here.



Worked example

Two basketballers' points in 5 games: Ana 12, 15, 14, 13, 16; Bo 5, 25, 8, 22, 10. Who is more consistent?

1. Both have mean 14
2. Ana range 4; Bo range 20

Answer: Ana (smaller range)

YOU DO

On your own: read the worked example, then try these.

- E1** Who would you pick for a game where you need at least 12 points? Why?

- E2** Make up two data sets with the same mean but very different ranges.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

What is a back-to-back stem-and-leaf plot used for?

Level 2

Group A range 12, group B range 30. Which is more consistent?

Level 3

Group A median 65, group B median 58. Which did better?

Lesson summary

Missed the lesson? Start here.

Remember

1. Put the two groups on the two sides of the stems.
2. Leaves get larger moving away from the stem.
3. Compare centre: mean or median.
4. Compare spread: a smaller range means more consistent.

Level 1

Highest mark

1. Largest leaf

Answer: 94, a girl

Level 2

Girls median

1. Mean of the 10th and 11th

Answer: 54

Level 3

Boys range

1. $93 - 10$

Answer: 83

Extension

Ana vs Bo

1. Same mean, compare range

Answer: Ana

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.